RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Chapter 9B: Applications of Integration

SYLLABUS INCLUDES

- Concept of definite integral as a limit of sum
- Definite integral as the area under a curve
- Evaluation of definite integrals
- Finding the area of region bounded by a curve and lines parallel to the coordinate axes, between a curve and a line, or between two curves
- Area below the x-axis
- Finding the volume of revolution about the x- or y-axis
- Finding the approximate value of a definite integral using a graphing calculator

PRE-REQUISITES

- Functions
- Differentiation Techniques
- Graphing Techniques
- Integration Techniques
- Small Angle Approximations
- Formula for Surface Area and Volume of basic shapes such as Sphere, Cylinder, Cone, Cuboid, etc.

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2 Volume of Solid of Revolution

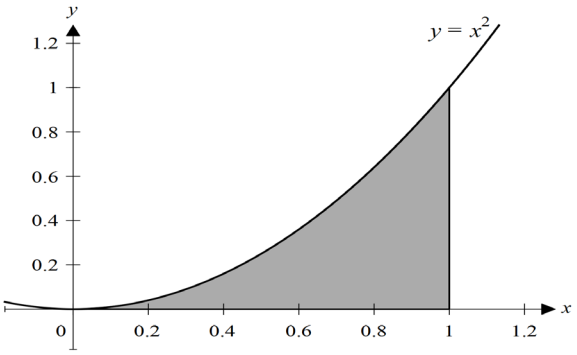
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1 Area under a Curve

Just as differentiation arises from the study of tangents, integral calculus is motivated by a geometric problem – the study of areas bounded by curves. Specifically, areas under graphs are meaningful representations of important related quantities. As it turns out, the processes of differentiation and integration are closely related.

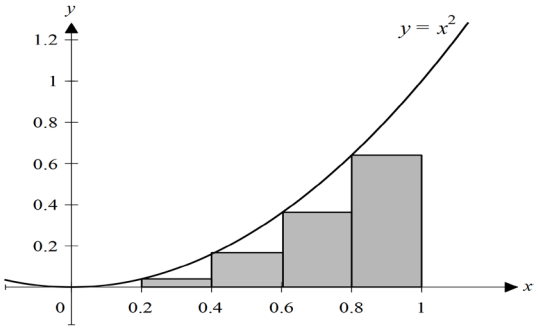
1.1 Numerical Integration − Archimedes' Problem

We begin our study of Integral Calculus by examining a problem faced by Archimedes over 2000 years ago − finding the exact area bounded by the graph of $y = x^2$, the x-axis and the lines $x = 0$ and $x = 1$.



Archimedes initially reasoned that this area could be approximated by rectangular intervals of equal width. Two such methods of the approximation are shown below.

Calculate the total area of the rectangles in each approximation below and indicate whether the approximated area is an under or over-estimate of the actual area.

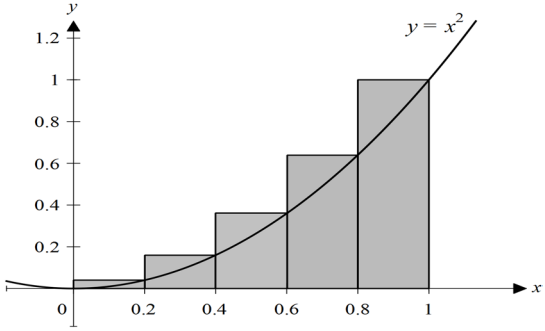


Interval width = **0.2 units**

x	0	0.2	0.4	0.6	0.8	1
y	0	0.04	0.16	0.36	0.64	1

Total area of rectangles = **0.24 units²**

This is an **under**-estimate of the area.



Interval width = **0.2 units**

x	0	0.2	0.4	0.6	0.8	1
y	0	0.04	0.16	0.36	0.64	1

Total area of rectangles = **0.44 units²**

This is an **over**-estimate of the area.

In fact, this technique of approximating areas is called numerical integration and is often useful in obtaining approximate areas that cannot be easily evaluated.

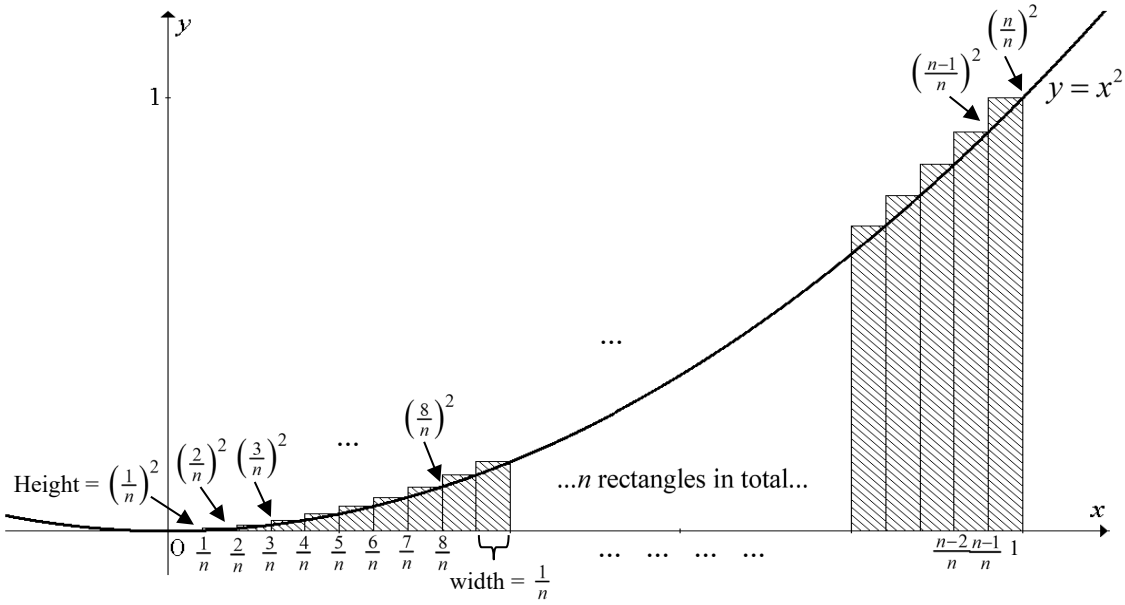
Observe

No. of Rectangles	Total Area of Rectangles	
	Under-estimate	Over-estimate
10	0.285	0.385
50	0.3234	0.3434
100	0.3284	0.3383

The table above lists out the total area of rectangles when carrying out the approximation with 10, 50 and 100 rectangles. Observe how the total area of rectangles varies as the number of rectangles used in the approximation increases. What can we conclude about the total area of rectangles as the number of intervals used increases?

Limiting Sums and the Definite Integral

Consider $y = x^2$ and we intend to find the area under the curve from $x = 0$ to $x = 1$. Divide the interval $[0,1]$ into *n* subintervals with equal width $\frac{1}{n}$ and we first approximate the area under the curve by *n* vertical rectangles (see diagram below).



The height of the 1st rectangle is $\left(\frac{1}{n}\right)^2$, the height of the 2nd rectangle is $\left(\frac{2}{n}\right)^2$ and so on.

Let the sum of the areas of *n* rectangles be A_n, then

$$A_n = \frac{1}{n}\times\left(\frac{1}{n}\right)^2 + \frac{1}{n}\times\left(\frac{2}{n}\right)^2 + \ldots + \frac{1}{n}\times\left(\frac{n-1}{n}\right)^2 + \frac{1}{n}\times\left(\frac{n}{n}\right)^2$$

$$= \frac{1}{n^3}\sum_{r=1}^{n} r^2$$

$$= \frac{1}{n^3}\times\frac{n(n+1)(2n+1)}{6}$$

$$= \frac{2n^2+3n+1}{6n^2}$$

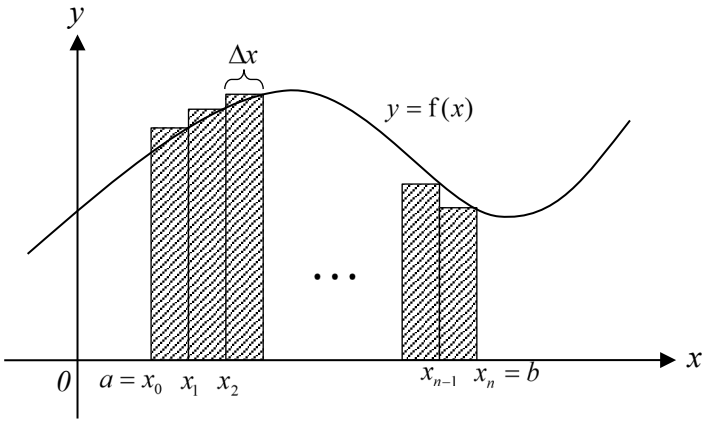
$$= \frac{1}{3}+\frac{1}{2n}+\frac{1}{6n^2}.$$

From the diagram, it can be observed that the approximation improves as the number of rectangles increases.

In fact, as $n \to \infty$, $\frac{1}{2n} \to 0$ and $\frac{1}{6n^2} \to 0$ and so we have $A_n \to \frac{1}{3}$.

This agrees with the "actual" area under the curve, i.e. $\int_0^1 x^2 \mathrm{d}x = \left[\frac{1}{3}x^3\right]_0^1 = \frac{1}{3}$.

In general, consider the graph of $y = \mathrm{f}(x)$.



Divide the interval $[a,b]$ into n subintervals with equal width $\Delta x = \dfrac{b-a}{n}$,

Let the n subintervals be $[x_0,x_1],[x_1,x_2],\ldots,[x_{n-1},x_n]$ where $x_0 = a$, $x_1 = a+\Delta x$, $x_2 = a+2\Delta x$, …, $x_{n-1} = a+(n-1)\Delta x$ and $x_n = a+n\Delta x = b$.

For i from 1 to n,
Width of i-th rectangle $= \Delta x$
Height of i-th rectangle $= \mathrm{f}(x_i)$

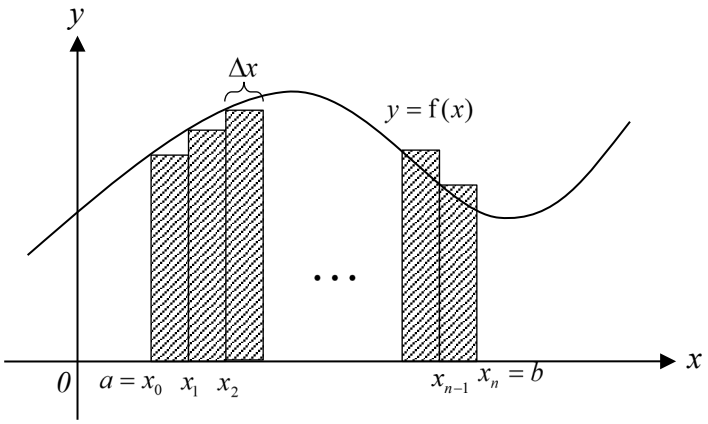
The sum of the areas of n rectangles, A_n

$$= \mathrm{f}(x_1)\Delta x + \mathrm{f}(x_2)\Delta x + \ldots + \mathrm{f}(x_n)\Delta x = \sum_{i=1}^{n} \mathrm{f}(x_i)\Delta x$$

The area under the curve of $y = \mathrm{f}(x)$ from $x = a$ to $x = b$ is the limit of the sum of the areas of approximating rectangles and we write

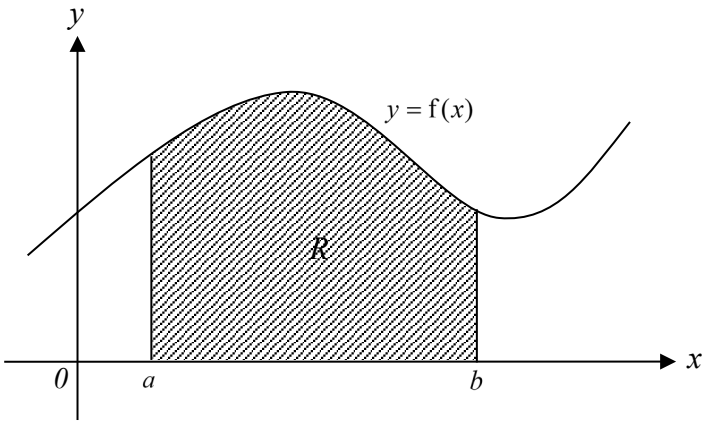
$$\int_a^b \mathrm{f}(x)\mathrm{d}x = \lim_{n\to\infty} \sum_{i=1}^{n} \mathrm{f}(x_i)\Delta x$$

In the diagram above, the height of the rectangle is taken to be the value of f at the right end-point of each interval. One can also use the left end-point for the approximating rectangles, i.e. $A_n = \mathrm{f}(x_0)\Delta x + \mathrm{f}(x_1)\Delta x + \ldots + \mathrm{f}(x_{n-1})\Delta x = \sum_{i=0}^{n-1} \mathrm{f}(x_i)\Delta x$ and $\int_a^b \mathrm{f}(x)\mathrm{d}x = \lim_{n\to\infty} \sum_{i=0}^{n-1} \mathrm{f}(x_i)\Delta x$.

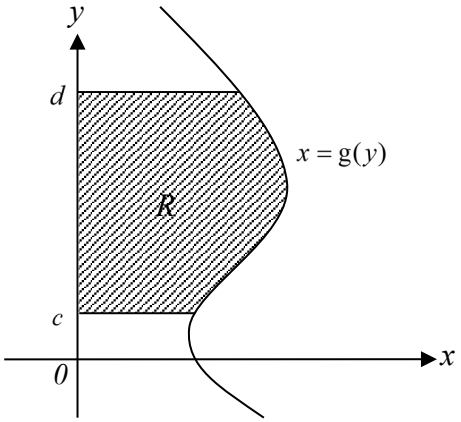


Area under curve as definite integral

Let R be the region bounded by the curve $y = \mathrm{f}(x)$, the x-axis and the lines $x = a$ and $x = b$ where $a \le b$ and $\mathrm{f}(x) \ge 0$ for $a \le x \le b$, then area of $R = \int_a^b \mathrm{f}(x)\,\mathrm{d}x$.



Similarly, if R is the region bounded by the curve $x = \mathrm{g}(y)$, the y-axis and the lines $y = c$ and $y = d$ where $c \le d$ and $\mathrm{g}(y) \ge 0$ for $c \le y \le d$, then area of $R = \int_c^d \mathrm{g}(y)\,\mathrm{d}y$.



1.2 The Fundamental Theorem of Calculus

Newton and Leibniz realised that the area under a curve of a function could be determined by a related function called the **integral** of the function.

The Fundamental Theorem of Calculus

Given that $\mathrm{F}(x)$ is any integral of $\mathrm{f}(x)$, i.e. $\dfrac{\mathrm{d}}{\mathrm{d}x}\mathrm{F}(x)=\mathrm{f}(x)$, then

$$\int_a^b \mathrm{f}(x)\,\mathrm{d}x=\mathrm{F}(b)-\mathrm{F}(a)\text{. (denoted by } \left[\mathrm{F}(x)\right]_a^b\text{)}$$

Note that the definite integral $\int_a^b \mathrm{f}(x)\,\mathrm{d}x$ is the area bounded by the curve $y=\mathrm{f}(x)$, the x-axis and the lines $x=a$ and $x=b$ where $a\le b$ and $\mathrm{f}\left(x\right)\ge 0$ for $a\le x\le b$. Essentially, the Fundamental Theorem of Calculus provides us with a quick method of evaluating this area, provided we are able to obtain F, an antiderivative of f.

Example 1

If $\mathrm{F}\left(x\right)=\ln\left(\dfrac{x^2-1}{x}\right)$ and $\dfrac{\mathrm{d}}{\mathrm{d}x}\mathrm{F}(x)=\mathrm{f}(x)$, find $\int_2^5 \mathrm{f}\left(x\right)\mathrm{d}x$, leaving your answer in exact form.

Solution:

By the Fundamental Theorem of Calculus,

$$\begin{aligned}\int_2^5 \mathrm{f}(x)\,\mathrm{d}x&=\left[\ln\left(\frac{x^2-1}{x}\right)\right]_2^5\\&=\ln\frac{24}{5}-\ln\frac{3}{2}\\&=\ln\frac{16}{5}\end{aligned}$$

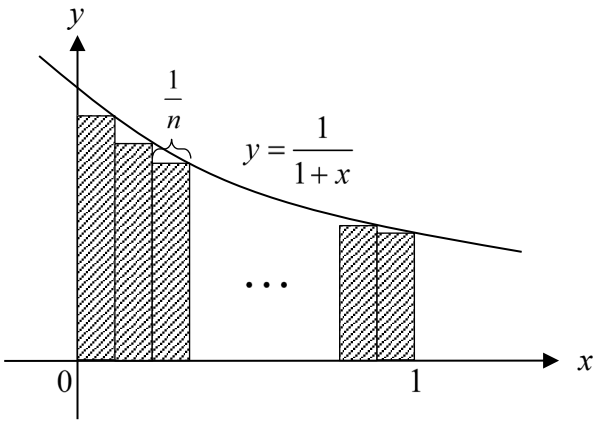
Example 2

The part of the graph of $y = \dfrac{1}{1+x}$ is shown in the diagram below and n rectangles, each of width $\dfrac{1}{n}$, are drawn under the curve between $x = 0$ and $x = 1$.

(i) Calculate the total area of the rectangles when $n = 5$, correct to 4 decimal places.

(ii) Show that the total area of the n rectangles can be expressed as $\displaystyle\sum_{r=1}^{n} \frac{1}{n+r}$.

(iii) Hence evaluate $\displaystyle\lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n+r}$.



Solution:

(i) Total area of 5 rectangles

$$= \frac{1}{5}\times\frac{1}{1+\frac{1}{5}} + \frac{1}{5}\times\frac{1}{1+\frac{2}{5}} + \frac{1}{5}\times\frac{1}{1+\frac{3}{5}} + \frac{1}{5}\times\frac{1}{1+\frac{4}{5}} + \frac{1}{5}\times\frac{1}{1+\frac{5}{5}} = 0.6456$$

(ii) Total area of n rectangles

$$= \frac{1}{n}\times\frac{1}{1+\frac{1}{n}} + \frac{1}{n}\times\frac{1}{1+\frac{2}{n}} + \ldots + \frac{1}{n}\times\frac{1}{1+\frac{n-1}{n}} + \frac{1}{n}\times\frac{1}{1+\frac{n}{n}}$$

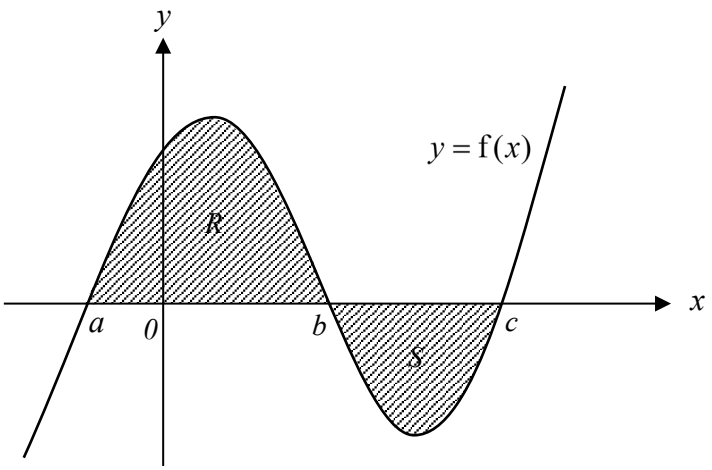
$$= \frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{n+n-1} + \frac{1}{n+n}$$

$$= \sum_{r=1}^{n} \frac{1}{n+r}$$

(iii) $\displaystyle\lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n+r} = \int_0^1 \frac{1}{1+x}\,\mathrm{d}x = \Big[\ln|1+x|\Big]_0^1 = \ln 2$

1.3 Evaluating Areas Bounded by Curves and Axes

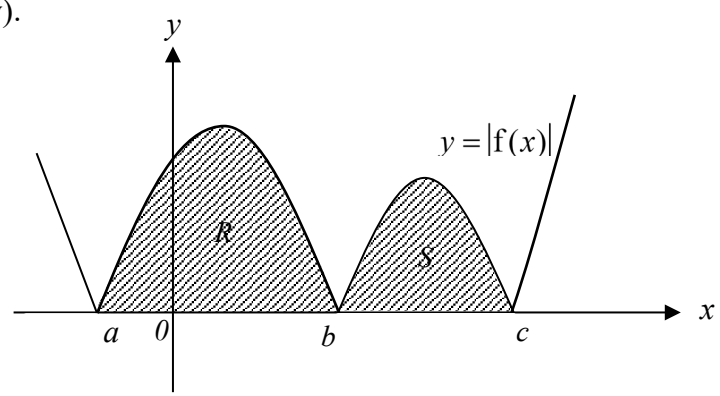
For curves **above** the x-axis, the definite integral is **positive**, whilst, for curves **below** the x-axis, the definite integral is **negative**.



1. Since $\mathrm{f}(x) \geq 0$ for all x in Region R, area of region R is $\int_a^b \mathrm{f}(x)\,\mathrm{d}x$.

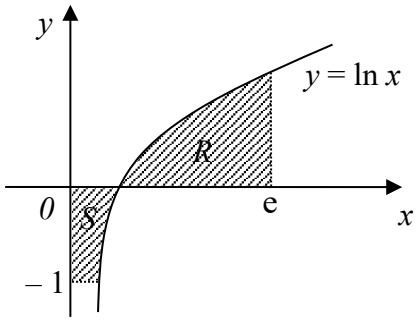
2. Since $\mathrm{f}(x) \leq 0$ for all x in Region S, area of region S is $-\int_b^c \mathrm{f}(x)\,\mathrm{d}x$ or $\int_b^c |\mathrm{f}(x)|\,\mathrm{d}x$.

3. Total area of region R and region S is $\int_a^b \mathrm{f}(x)\,\mathrm{d}x + \left(-\int_b^c \mathrm{f}(x)\,\mathrm{d}x\right)$ or just $\int_a^c |\mathrm{f}(x)|\,\mathrm{d}x$ (see diagram below).



Example 3

The figure below shows the curve $y = \ln x$. Calculate the total areas of the regions R and S.



Solution:

Total areas of R and $S = \mathrm{e}^{-1} + \int_{\mathrm{e}^{-1}}^{\mathrm{e}} |\ln x|\,\mathrm{d}x$

or $\mathrm{e}^{-1} - \int_{\mathrm{e}^{-1}}^{1} \ln x\,\mathrm{d}x + \int_1^{\mathrm{e}} \ln x\,\mathrm{d}x$

$= 1.63$ units^2 (3 s.f.) or $2 - \mathrm{e}^{-1}$ units^2

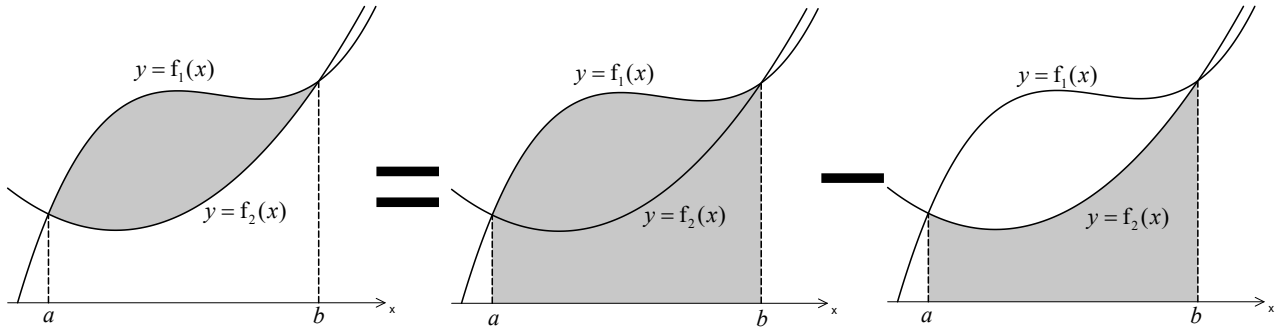
Alternatively: Total areas of R and S

$= \left[\mathrm{e} - \int_0^1 \mathrm{e}^y\,\mathrm{d}y\right] + \int_{-1}^0 \mathrm{e}^y\,\mathrm{d}y$

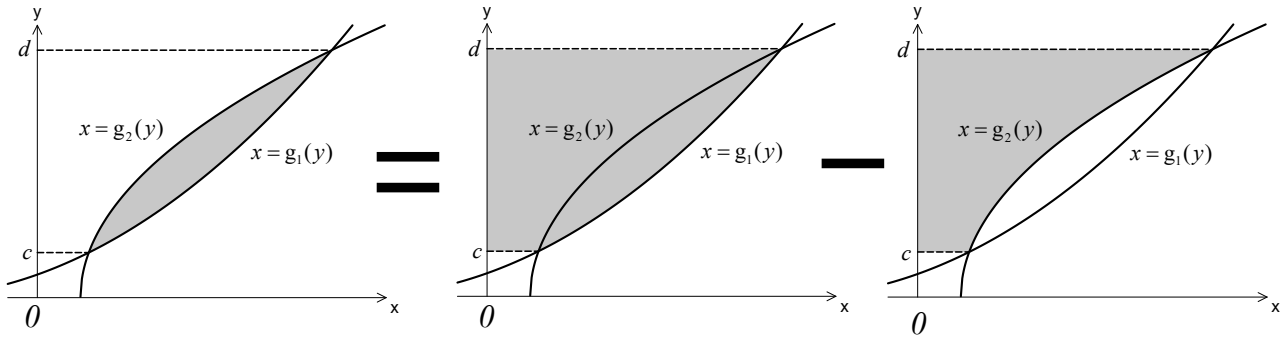
1.4 Evaluating Areas Between Curves

In this section, we will be discussing the evaluation of areas between two graphs.

Consider the curves $y = \mathrm{f}_1(x)$ and $y = \mathrm{f}_2(x)$, and they intersect at $x = a$ and $x = b$, as shown below. Then, the area bounded by the two curves between $x = a$ and $x = b$ can be found by finding the difference between the area under $y = \mathrm{f}_1(x)$ and the area under $y = \mathrm{f}_2(x)$ between $x = a$ and $x = b$.



In general, given 2 curves defined by the equations $y = \mathrm{f}_1(x)$ and $y = \mathrm{f}_2(x)$, and an interval $[a, b]$ where $\mathrm{f}_1(x) \geq \mathrm{f}_2(x)$ within this interval, we can evaluate the area bounded between them from $x = a$ to $x = b$ by evaluating the definite integral $\int_a^b (\mathrm{f}_1(x) - \mathrm{f}_2(x))\,\mathrm{d}x$, or we can also use $\int_a^b |\mathrm{f}_1(x) - \mathrm{f}_2(x)|\,\mathrm{d}x$.

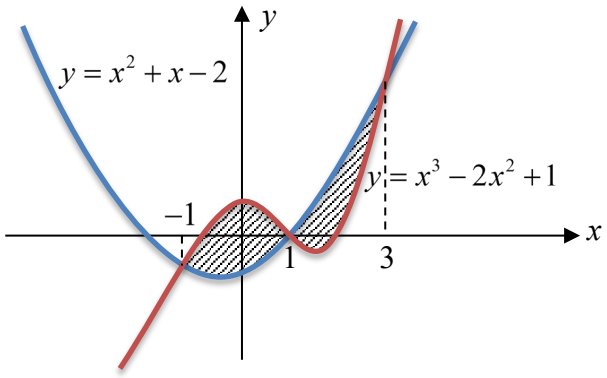


Similarly, given 2 curves defined by the equations $x = \mathrm{g}_1(y)$ and $x = \mathrm{g}_2(y)$, and an interval $[c, d]$ where $\mathrm{g}_1(y) \geq \mathrm{g}_2(y)$ within this interval, we can evaluate the area bounded between them from $y = c$ to $y = d$ by evaluating the definite integral $\int_c^d (\mathrm{g}_1(y) - \mathrm{g}_2(y))\,\mathrm{d}y$, or we can also use $\int_c^d |\mathrm{g}_1(y) - \mathrm{g}_2(y)|\,\mathrm{d}y$

Example 4

Find the area bounded by the curves of $y = x^3 - 2x^2 + 1$ and $y = x^2 + x - 2$ between $x = -1$ and $x = 3$.

Solution:



From the GC, the two graphs intersect at $x = -1,\ 1$ and 3.

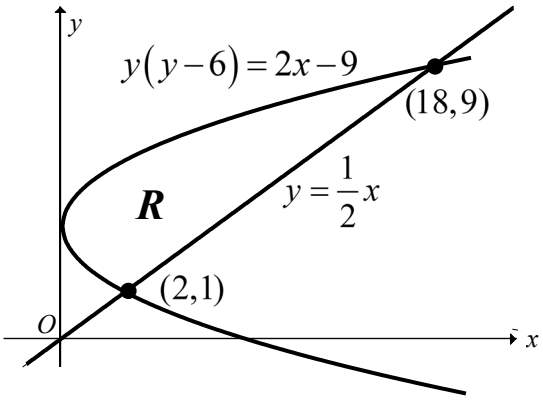
Required area $= \int_{-1}^{1} (x^3 - 2x^2 + 1) - (x^2 + x - 2)\ \mathrm{d}x + \int_{1}^{3} (x^2 + x - 2) - (x^3 - 2x^2 + 1)\ \mathrm{d}x$

$= 8$ units^2

Alternatively: Required area $= \int_{-1}^{3} \left|(x^3 - 2x^2 + 1) - (x^2 + x - 2)\right|\ \mathrm{d}x = 8$ units^2

Example 5

The diagram (not drawn to scale) shows the region R which is bounded by the curve $y(y-6) = 2x - 9$ and the line $y = \frac{1}{2}x$. Find the exact area of R.



Solution:

$y(y-6) = 2x - 9 \quad \Rightarrow \quad x = \dfrac{y(y-6)+9}{2}$

Area of $R = \int_{1}^{9} 2y - \dfrac{y(y-6)+9}{2}\ \mathrm{d}y$

$= \int_{1}^{9} -\frac{1}{2}y^2 + 5y - \frac{9}{2}\ \mathrm{d}y$

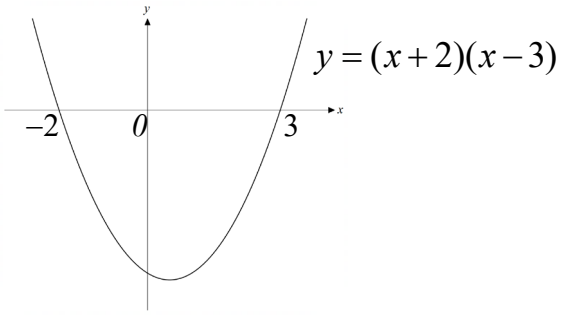
$= \left[-\frac{1}{6}y^3 + \frac{5}{2}y^2 - \frac{9}{2}y\right]_1^9 = \left(-\frac{729}{6} + \frac{405}{2} - \frac{81}{2}\right) - \left(-\frac{1}{6} + \frac{5}{2} - \frac{9}{2}\right) = \frac{128}{3}$ units^2

Evaluating definite integrals of $|\,f(x)\,|$

Example 6

Solve exactly $\int_{-1}^{4}\left|(x+2)(x-3)\right|\,\mathrm{d}x$.

Solution:



Consider $(x+2)(x-3)<0$, we have $-2<x<3$

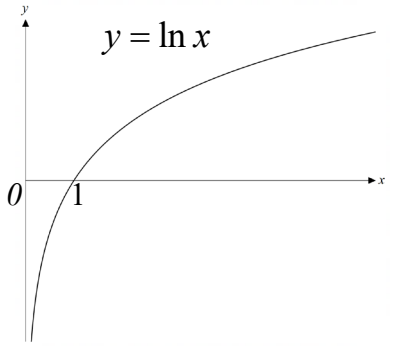
$$\begin{aligned}\int_{-1}^{4}\left|(x+2)(x-3)\right|\,\mathrm{d}x &= \int_{-1}^{3}-(x+2)(x-3)\,\mathrm{d}x+\int_{3}^{4}(x+2)(x-3)\,\mathrm{d}x\\ &= -\int_{-1}^{3}x^2-x-6\,\mathrm{d}x+\int_{3}^{4}x^2-x-6\,\mathrm{d}x\\ &= -\left[\frac{x^3}{3}-\frac{x^2}{2}-6x\right]_{-1}^{3}+\left[\frac{x^3}{3}-\frac{x^2}{2}-6x\right]_{3}^{4}\\ &= 21.5\end{aligned}$$

Example 7

Solve exactly $\int_{0.5}^{\mathrm{e}}\left|\ln x\right|\,\mathrm{d}x$

Solution:

Consider $\ln x>0$, we have $0<x<1$



$$\begin{aligned}\int_{0.5}^{\mathrm{e}}\left|\ln x\right|\,\mathrm{d}x &= -\int_{0.5}^{1}\ln x\,\mathrm{d}x+\int_{1}^{\mathrm{e}}\ln x\,\mathrm{d}x\\ &= -\left(\left[x\ln x\right]_{0.5}^{1}-\int_{0.5}^{1}1\,\mathrm{d}x\right)+\left(\left[x\ln x\right]_{1}^{\mathrm{e}}-\int_{1}^{\mathrm{e}}1\,\mathrm{d}x\right)\\ &= -(-0.5\ln 0.5-0.5)+(\mathrm{e}-\mathrm{e}+1)\\ &= 0.5\ln 0.5+1.5\end{aligned}$$

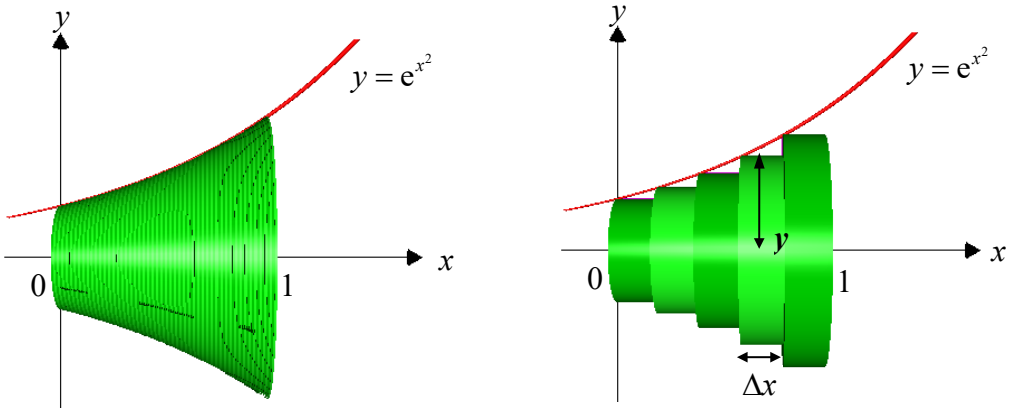
2 Volume of Solid of Revolution (Disc Method)

The solid formed when a finite region formed by part of a curve is rotated about a straight line is called a solid of revolution.

Consider the graph defined by the equation $y = \mathrm{e}^{x^2}$. What is the volume of the solid of revolution formed when the finite region bounded by the curve, the x-axis and the lines $x = 0$ and $x = 1$ is rotated completely about a horizontal or vertical line? We have seen that the area under a curve can be expressed as the limit of a sum of areas of rectangles. Can we apply a similar idea for volume?

2.1 Rotation about the x-axis ($y = 0$)

Now consider the finite region bounded by the curve $y = \mathrm{e}^{x^2}$, the axes and the line $x = 1$ and rotate it completely about the x-axis, to give a solid of revolution, as shown below.



To estimate the volume of this solid, we can divide the solid into n cylindrical sections that is perpendicular to the axis of rotation, and sum up the volume of these cylindrical sections. Since any cross-section perpendicular to the axis is circular, each section is approximately a cylindrical disc if the cuts are reasonably close.

Radius of i-th disc = y_i where $i = 1, 2, \ldots, n$

Area of cross-section of i-th disc = πy_i^2

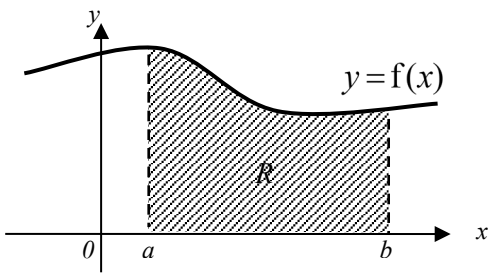
Volume of disc = $\pi y_i^2 \Delta x$

Estimated volume of solid = $\sum_{i=1}^{n} \pi y_i^2 \Delta x$

Exact volume of solid = $\lim_{n\to\infty} \sum_{i=1}^{n} \pi y_i^2 \Delta x = \int_0^1 \pi y^2 \, \mathrm{d}x = \int_0^1 \pi \left(\mathrm{e}^{x^2}\right)^2 \mathrm{d}x$ (≈ 7.43 units^3)

In general, if the Region R is bounded by the curve $y = \mathrm{f}(x)$, the x-axis, the lines $x = a$ and $x = b$, as shown on the right, then the volume of solid of revolution formed when R is rotated completely about the $x-$axis

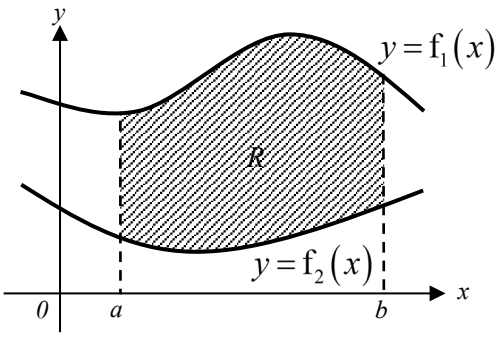
$= \pi \int_a^b y^2 \, \mathrm{d}x = \pi \int_a^b \left[\mathrm{f}(x)\right]^2 \mathrm{d}x$.



Furthermore, if the Region R is bounded by the curves $y = \mathrm{f}_1(x)$, $y = \mathrm{f}_2(x)$, the lines $x = a$ and $x = b$, as shown on the right, then the volume of solid of revolution formed when R is rotated completely about the x–axis

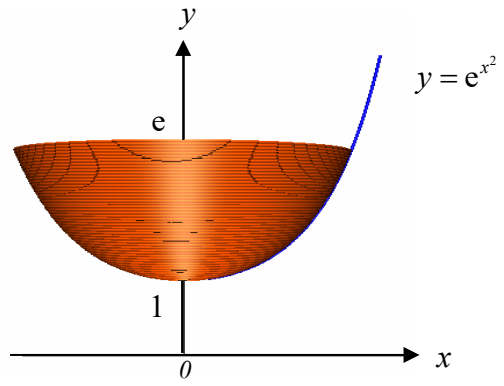
$$= \pi \int_a^b \left[\mathrm{f}_1(x)\right]^2 \,\mathrm{d}x - \pi \int_a^b \left[\mathrm{f}_2(x)\right]^2 \,\mathrm{d}x$$

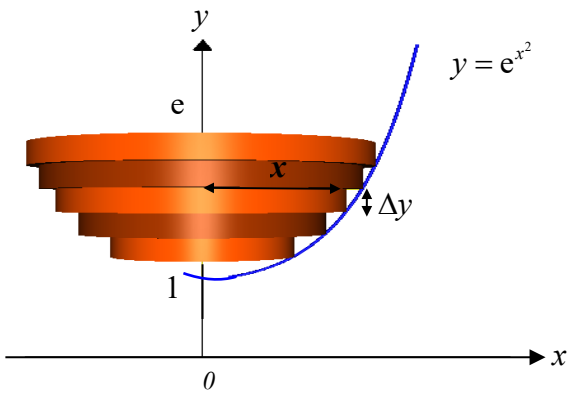
$$= \pi \int_a^b \left[\mathrm{f}_1(x)\right]^2 - \left[\mathrm{f}_2(x)\right]^2 \,\mathrm{d}x.$$



2.2 Rotation about the *y*-axis (*x* = 0)

Now consider the finite region bounded by the curve $y = \mathrm{e}^{x^2}$, the y-axis and the line $y = \mathrm{e}$ and rotate it completely about the y-axis, to give a solid of revolution, as shown below.



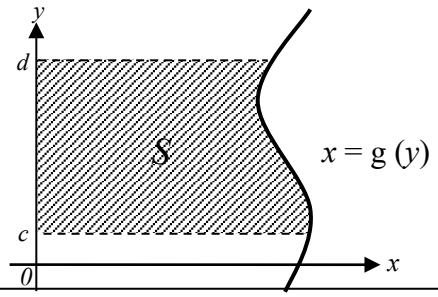


A similar argument gives

$$\text{Volume} = \lim_{n\to\infty} \sum_{i=1}^{n} \pi x_i^2 \Delta y = \int_1^{\mathrm{e}} \pi x^2 \,\mathrm{d}y = \int_1^{\mathrm{e}} \pi \left(\sqrt{\ln y}\right)^2 \mathrm{d}y = \pi \text{ units}^3$$

In general, if the Region S is bounded by the curve $x = \mathrm{g}(y)$, the y-axis, the lines $y = c$ and $y = d$, as shown on the right, then the volume of solid of revolution formed when S is rotated completely about the y–axis

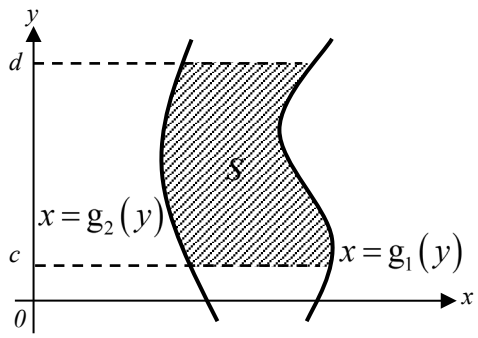
$$= \pi \int_c^d x^2 \,\mathrm{d}y = \pi \int_c^d \left[\mathrm{g}(y)\right]^2 \,\mathrm{d}y$$



Furthermore, if the Region S is bounded by the curves $x = \mathrm{g}_1(y)$, $x = \mathrm{g}_2(y)$, the lines $y = c$ and $y = d$, as shown on the right, then the volume of solid of revolution formed when S is rotated completely about the y–axis

$$= \pi \int_c^d \left[\mathrm{g}_1(y)\right]^2 \,\mathrm{d}y - \pi \int_c^d \left[\mathrm{g}_2(y)\right]^2 \,\mathrm{d}y$$

$$= \pi \int_c^d \left[\mathrm{g}_1(y)\right]^2 - \left[\mathrm{g}_2(y)\right]^2 \,\mathrm{d}y.$$



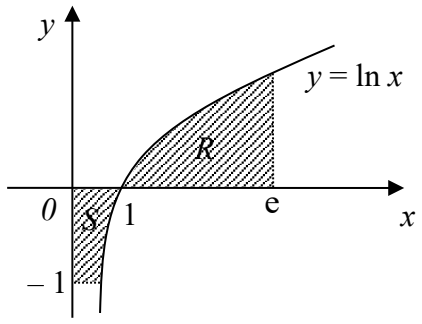
2.3 Examples on Solid of Revolution

Example 8

The figure below shows the curve $y = \ln x$ and the two regions R and S. Find the volume of the solid of revolution obtained when

(a) R is rotated 2π radians about the x-axis;

(b) S is rotated 2π radians about the x-axis.



Solution:

(a) Required volume $= \pi\int_1^{\mathrm{e}} y^2 \,\mathrm{d}x$

$= \pi\int_1^{\mathrm{e}} (\ln x)^2 \,\mathrm{d}x$

$= 2.26 \text{ units}^3$ (3 s.f.)

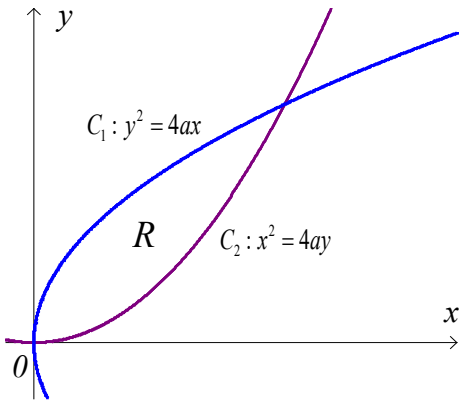
(b) $y = -1 \Rightarrow x = \mathrm{e}^{-1}$

Required volume $= \pi(1)^2\mathrm{e}^{-1} + \pi\int_{\mathrm{e}^{-1}}^1 (\ln x)^2 \,\mathrm{d}x$

$= 1.66 \text{ units}^3$ (3 s.f.)

Example 9

The region R is bounded by the curves C_1 and C_2 with equations given by $y^2 = 4ax$ and $x^2 = 4ay$ respectively where $a > 0$, as shown on the right.



The curves C_1 and C_2 intersect at the points (0, 0) and ($4a$, $4a$). The region R is rotated through four right angles about the x-axis. Show that the volume of the solid obtained is $\dfrac{96\pi a^3}{5}$ units^3.

Solution:

Required volume $= \pi\int_0^{4a} \left(2\sqrt{ax}\right)^2 - \left(\frac{1}{4a}x^2\right)^2 \,\mathrm{d}x$

$= \pi\int_0^{4a} 4ax - \frac{x^4}{16a^2} \,\mathrm{d}x$

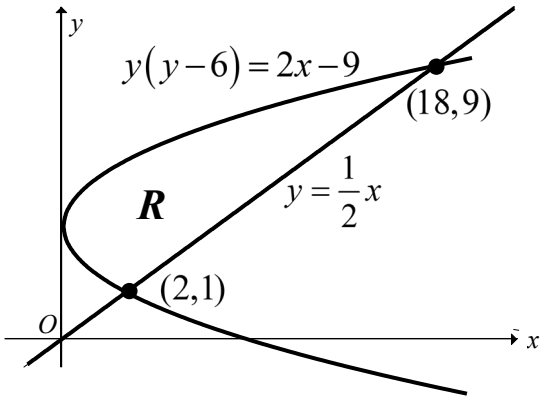
$= \pi\left[\frac{4ax^2}{2} - \frac{x^5}{80a^2}\right]_0^{4a}$

$= \pi\left[32a^3 - \frac{64}{5}a^3\right] = \frac{96}{5}a^3\pi \text{ units}^3$. (shown)

Example 10

The diagram (not drawn to scale) shows the region R which is bounded by the curve $y(y-6)=2x-9$ and the line $y=\frac{1}{2}x$.

Find the volume of the solid of revolution formed when R is rotated 360° about the x-axis, giving your answer to 1 decimal place.



Solution:

$$y(y-6)=2x-9$$
$$(y-3)^2-9=2x-9$$
$$y=3\pm\sqrt{2x}$$

Required volume $= \pi\int_0^{18}\left(3+\sqrt{2x}\right)^2\,\mathrm{d}x-\pi\int_0^{2}\left(3-\sqrt{2x}\right)^2\,\mathrm{d}x-\pi\int_2^{18}\left(\frac{1}{2}x\right)^2\,\mathrm{d}x$

$= 1340.4$ units^3 (1 d.p.)

Example 11

The region R is bounded by the curve $y=\sqrt[4]{1+x^2}$, the lines $x=0$ and $x=3$ and the x-axis.

The region S is bounded by the curve $y=\sqrt[4]{1+x^2}$, the line $y=\sqrt[4]{10}$ and the y-axis.

(i) Find the exact volume of the solid formed when S is rotated 2π radians about the y-axis.

(ii) Find the exact volume of the solid formed when R is rotated 2π radians about the y-axis.

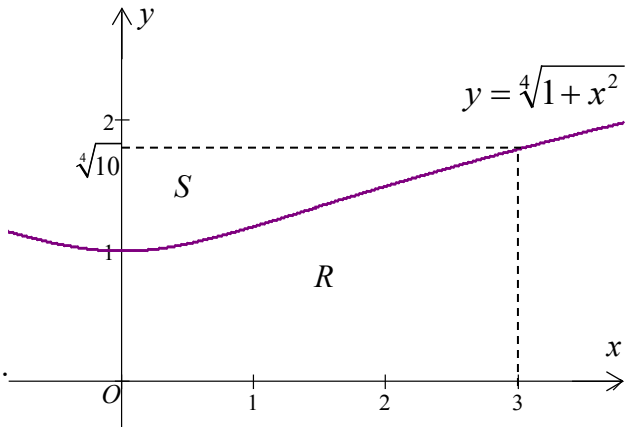
Solution:

(i) Required volume $= \pi\int_1^{\sqrt[4]{10}} x^2\,\mathrm{d}y$

$= \pi\int_1^{\sqrt[4]{10}} (y^4-1)\,\mathrm{d}y$

$= \pi\left[\frac{1}{5}y^5-y\right]_1^{\sqrt[4]{10}}$

$= (\sqrt[4]{10}+\frac{4}{5})\pi$ units^3.



(ii) Required volume $= \pi(3)^2\sqrt[4]{10}-\left(\sqrt[4]{10}+\frac{4}{5}\right)\pi$

$= \left[8\left(\sqrt[4]{10}\right)-\frac{4}{5}\right]\pi$ units^3.

Example 12

The diagram shows the curve $y=\frac{1}{2}\left(9-x^2\right)$. Find the exact volume of the solid of revolution obtained when the region bounded by the curve and the x-axis is rotated

(a) π radians about the y-axis;

(b) 2π radians about the line $x=-3$.

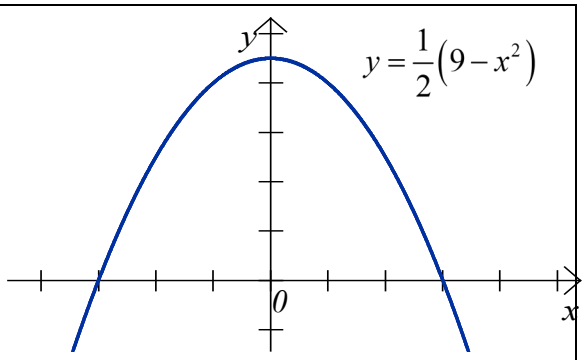
Solution:

(a) Required volume $= \pi\int_0^{9/2} x^2 \, dy$

$$= \pi\int_0^{9/2} (9-2y) \, dy$$

$$= \pi\left[9y-y^2\right]_0^{9/2}$$

$$= \frac{81}{4}\pi \text{ units}^3$$



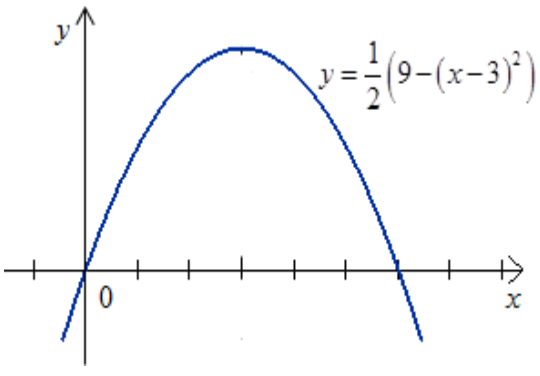
(b) Note: When the required region is rotated through 2π about other vertical or horizontal lines instead of one of the axes, we will need to rewrite the problem (usually via translation of the graph) so that the axis of rotation is one of the x or y-axis.

Translate the graph of $y=\frac{1}{2}\left(9-x^2\right)$ by 3 units in the positive x-direction.

The resulting graph has equation $y=\frac{1}{2}\left(9-(x-3)^2\right)$.

The required volume is the **same** as the volume of the solid obtained when the region bounded by $y=\frac{1}{2}\left(9-(x-3)^2\right)$ and the x-axis is rotated 2π radians about the y-axis.

$y=\frac{1}{2}\left(9-(x-3)^2\right) \Rightarrow x=3\pm\sqrt{9-2y}$



Required volume

$$= \pi\int_0^{9/2} \left(3+\sqrt{9-2y}\right)^2-\left(3-\sqrt{9-2y}\right)^2 \, dy$$

$$= \pi\int_0^{9/2} 12\sqrt{9-2y} \, dy$$

$$= \pi\left[-4(9-2y)^{3/2}\right]_0^{9/2} = 108\pi \text{ units}^3$$

SUMMARY